CSC3002F - Operating Systems II

Scheduling Assignment

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Introduction

Allegra the barman is an abstraction of the low-level workings of a CPU. The study of CPU scheduling algorithms can be likened much to the analogy of a barman serving drinks at a bar, where the overall goal is to most efficiently and effectively serve all customers such that the majority of customers will be satisfied with their experience.

The goal of this assignment is to determine the CPU scheduling algorithm that, when applied to the bar setting, maximises satisfaction for the most customers.

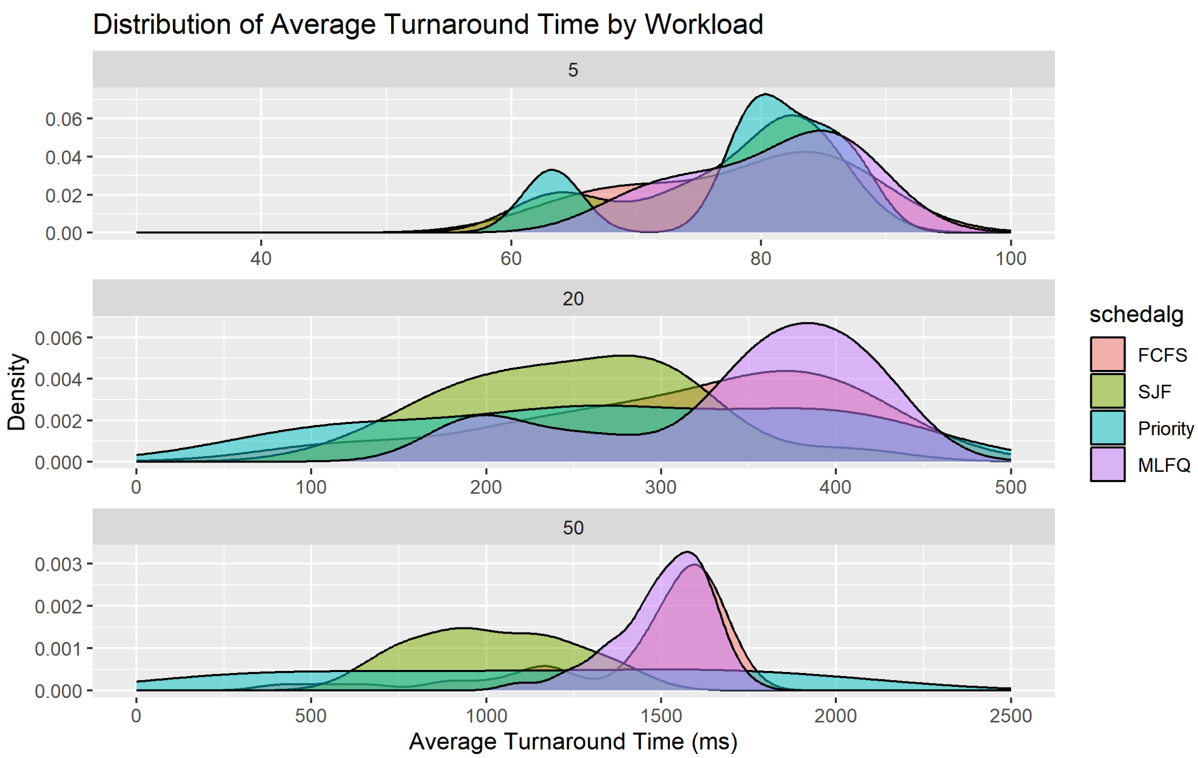
Methods

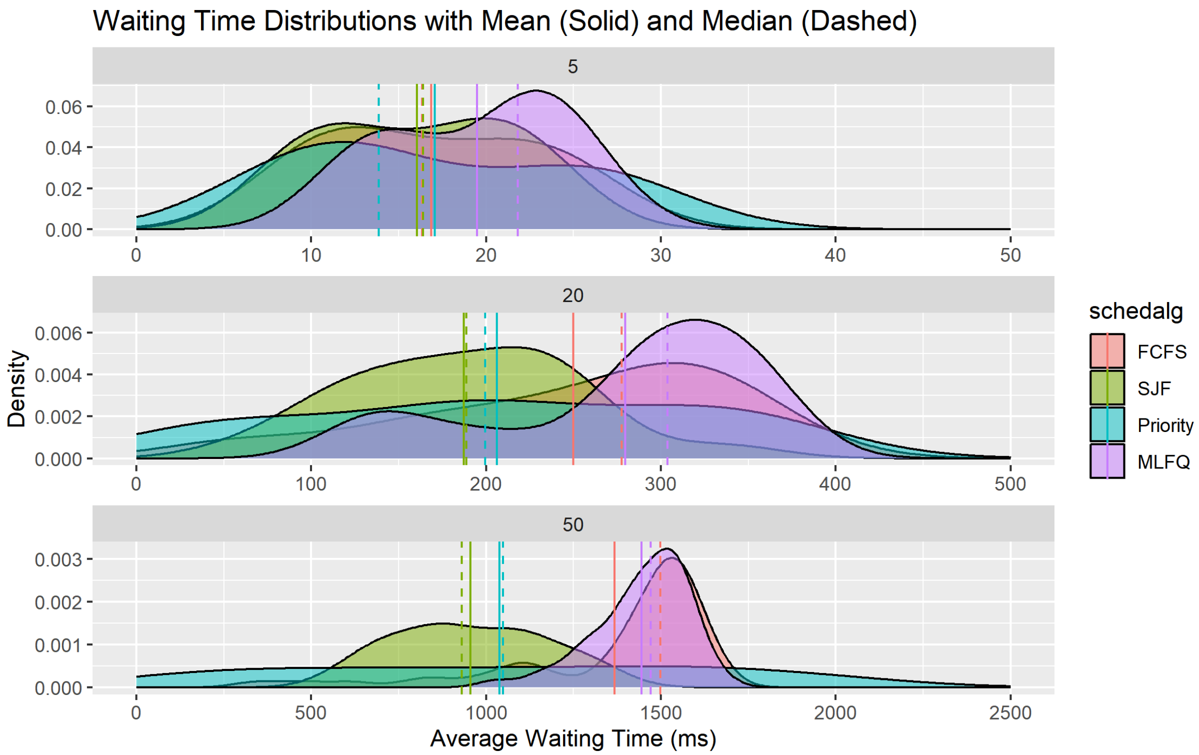
To generate the data

Results

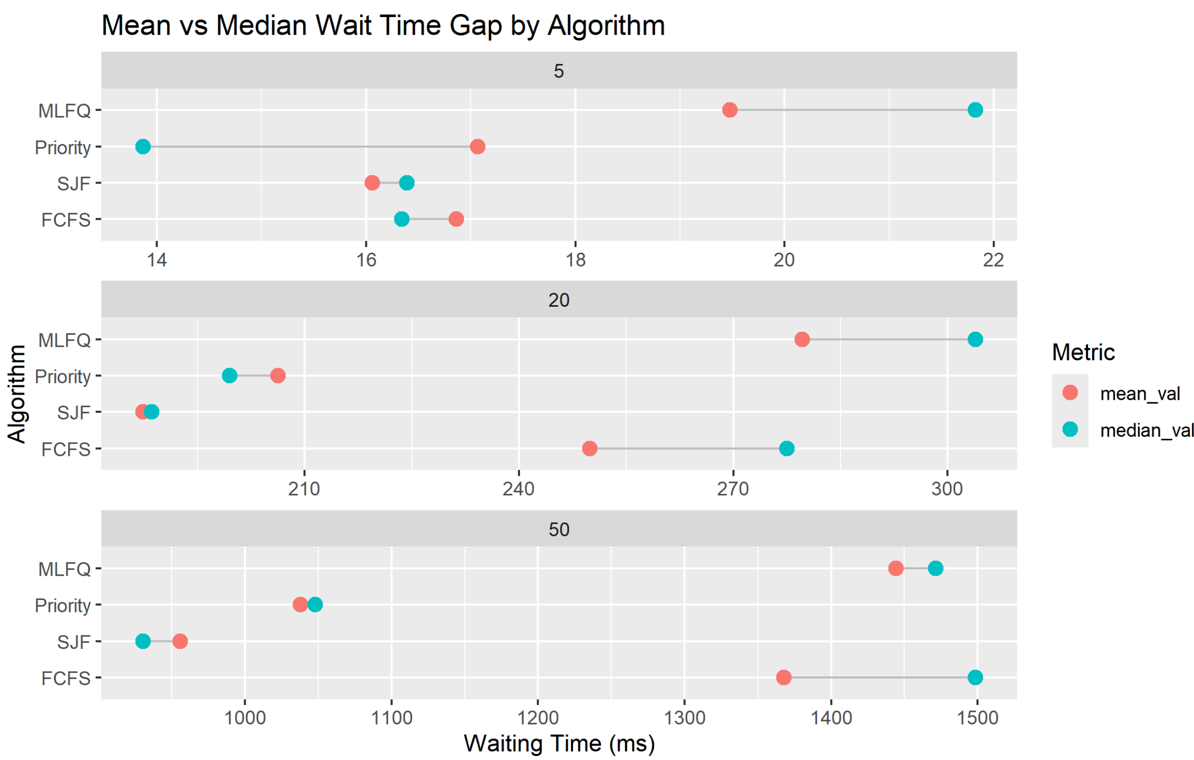
Distribution of Average Response Time by Workload
5
0.06
0.04
0.02
0.00
0
10
20
30
40
50
20
0.006
0.004
0.002
0.000
0
100
200
300
400
500
50
0.003
0.002
0.001
0.000
0
500
1000
1500
2000
2500
Density
Average Response Time (ms)
schedalg
FCFS
SJF
Priority
MLFQ


Distribution of Average Waiting Time by Workload
5
20
50
Density
Average Waiting Time (ms)
schedalg
FCFS
SJF
Priority
MLFQ


Distribution of Average Turnaround Time by Workload
5
20
50
Density
Average Turnaround Time (ms)
schedalg
FCFS
SJF
Priority
MLFQ

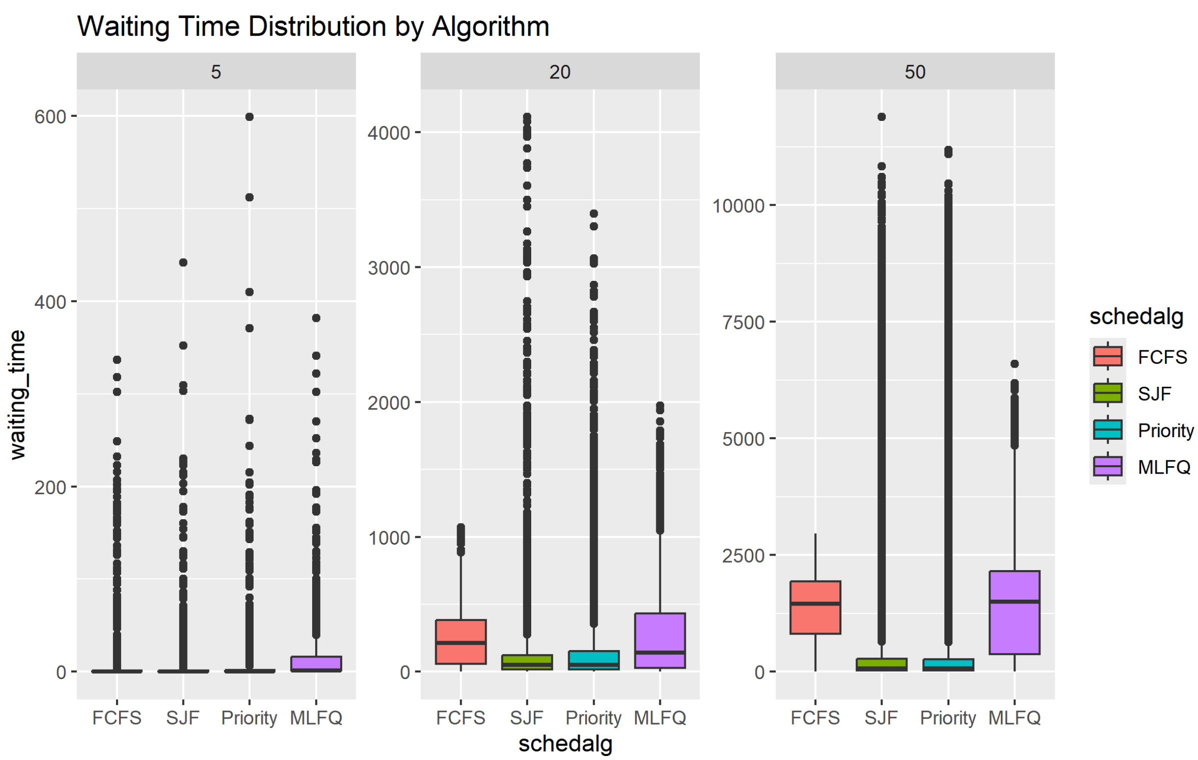


Waiting Time Distributions with Mean (Solid) and Median (Dashed)
5
20
50
Density
Average Waiting Time (ms)
schedalg
FCFS
SJF
Priority
MLFQ



Mean vs Median Wait Time Gap by Algorithm
5
20
50
MLFQ
Priority
SJF
FCFS
Algorithm
Waiting Time (ms)
Metric
mean_val
median_val

```r
ggplot(df, aes(x=schedalg, y=waiting_time, fill=schedalg)) +
  geom_boxplot() +
  facet_wrap(~total_patrons, scales = "free") +
  labs(title="Waiting Time Distribution by Algorithm")
```

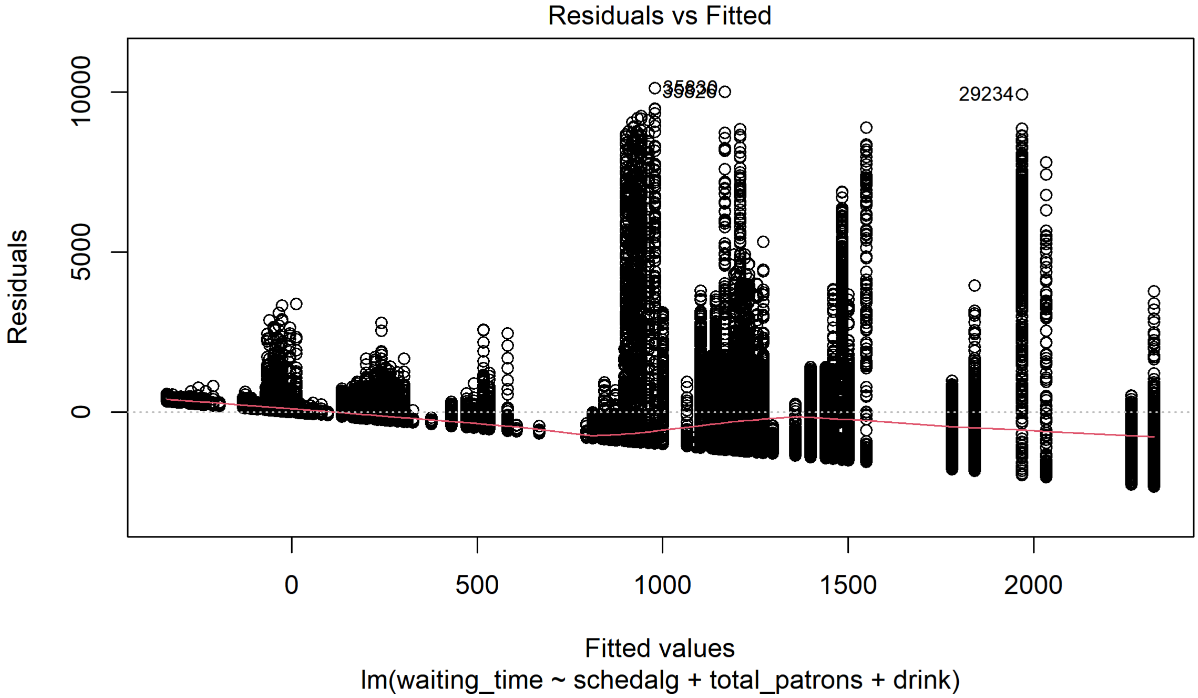



```r
# 1. Fit the model
fit <- lm(waiting_time ~ schedalg + total_patrons + drink, data = df)

# 2. View the significance (p-values) and magnitudes (coefficients)
summary(fit)
```

```
Call:
lm(formula = waiting_time ~ schedalg + total_patrons + drink,
    data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-2325.0  -833.6  -155.8   297.6 10117.2
```

```
Coefficients:
                   Estimate Std. Error t value Pr(>|t|)    
(Intercept)         1090.57      37.88  28.788  < 2e-16 ***
schedalgSJF         -295.37      18.80 -15.711  < 2e-16 ***
schedalgPriority    -230.18      18.80 -12.244  < 2e-16 ***
schedalgMLFQ          61.31      18.80   3.261  0.00111 ** 
total_patrons20      206.03      29.20   7.057 1.73e-12 ***
total_patrons50     1174.16      27.38  42.883  < 2e-16 ***
drinkBeer          -1104.22      36.60 -30.172  < 2e-16 ***
drinkBloody Mary   -1071.83      36.71 -29.197  < 2e-16 ***
drinkCider         -1054.76      35.58 -29.648  < 2e-16 ***
drinkCosmopolitan  -1126.20      37.48 -30.045  < 2e-16 ***
drinkDraft Beer    -1132.50      36.22 -31.269  < 2e-16 ***
drinkGin and Tonic -1111.39      36.48 -30.462  < 2e-16 ***
drinkMargarita      -866.04      36.60 -23.664  < 2e-16 ***
drinkMartini       -1115.89      36.04 -30.967  < 2e-16 ***
drinkMojito         -824.73      35.86 -22.997  < 2e-16 ***
drinkPina Colada    -484.36      36.00 -13.455  < 2e-16 ***
drinkRed Wine      -1092.50      35.73 -30.577  < 2e-16 ***
drinkTequila shot  -1132.41      37.16 -30.473  < 2e-16 ***
drinkVodka shot    -1094.05      36.16 -30.253  < 2e-16 ***
drinkWhite Wine    -1101.26      36.71 -29.998  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1339 on 40592 degrees of freedom
Multiple R-squared:  0.1611,	Adjusted R-squared:  0.1607 
F-statistic: 410.2 on 19 and 40592 DF,  p-value: < 2.2e-16
```

```r
# 3. Check for Fairness/Starvation by plotting residuals
plot(fit, which = 1)
```

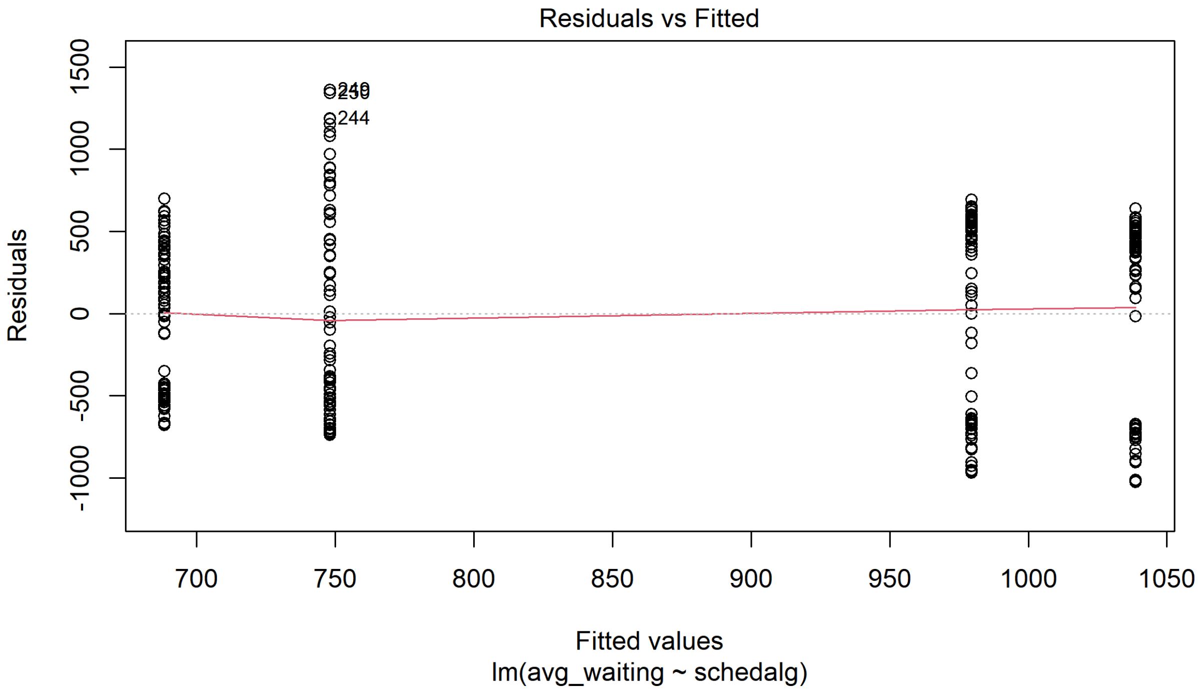


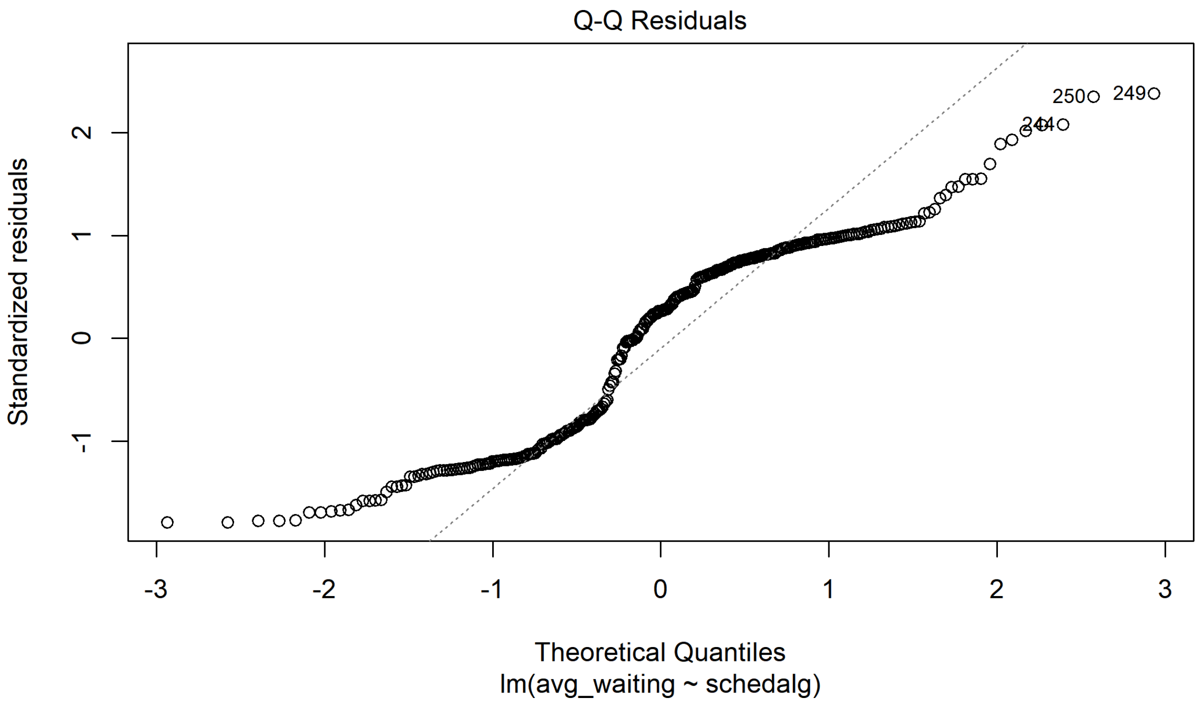

```r
# Compare means
model_speed <- lm(avg_waiting ~ schedalg, data = averages)
summary(model_speed)
```

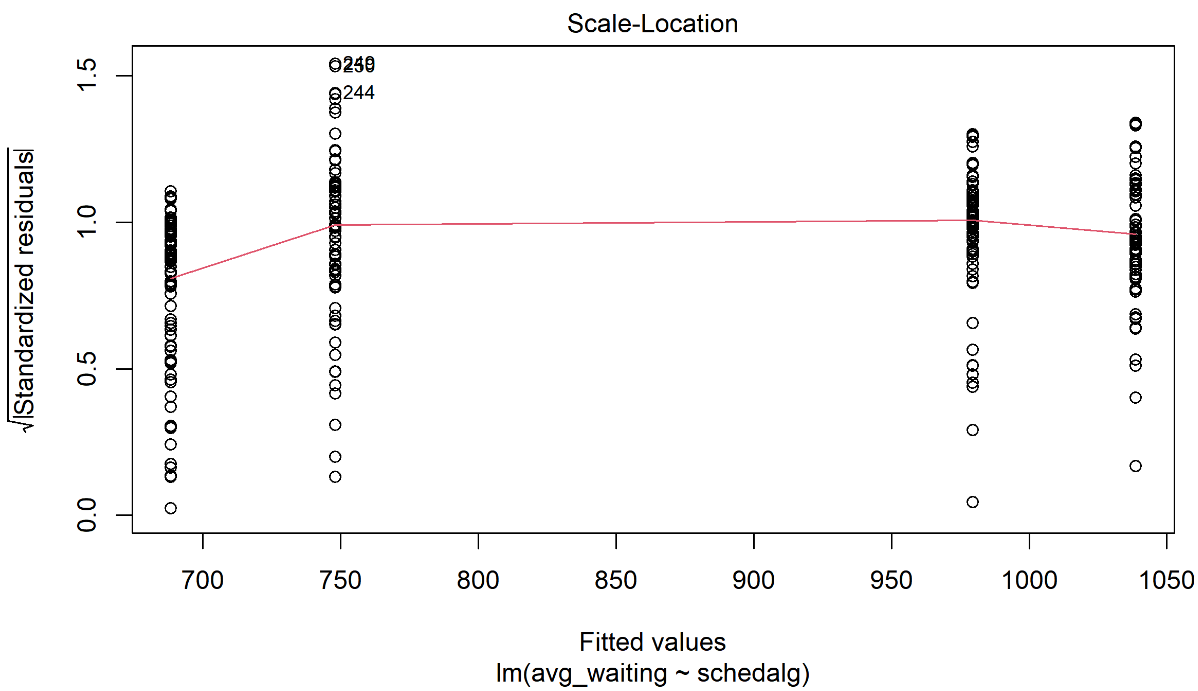
```
Call:
lm(formula = avg_waiting ~ schedalg, data = averages)

Residuals:
    Min      1Q  Median      3Q     Max
-1025.7  -580.0   149.7   472.0  1360.3

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)        979.61      66.53  14.724  < 2e-16 ***
schedalgSJF       -291.32      94.09  -3.096  0.00215 **
schedalgPriority  -231.54      94.09  -2.461  0.01443 *
schedalgMLFQ        59.18      94.09   0.629  0.52986
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 576.2 on 296 degrees of freedom
```

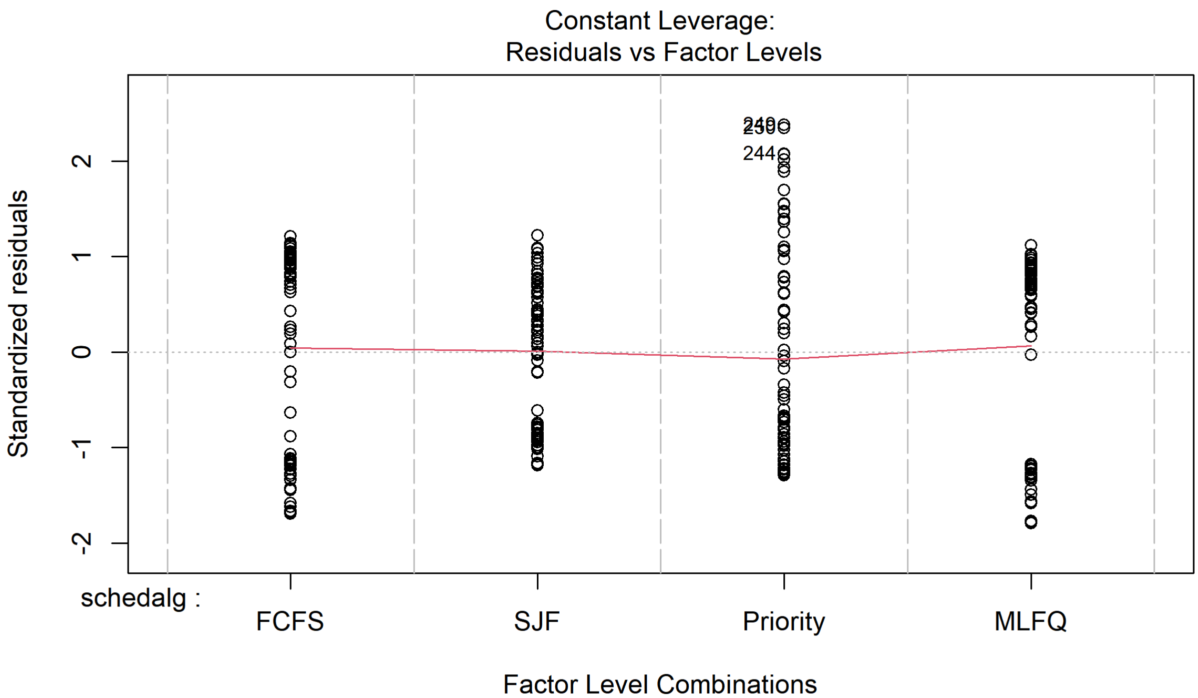
```
Multiple R-squared:  0.06309,	Adjusted R-squared:  0.0536 
F-statistic: 6.644 on 3 and 296 DF,  p-value: 0.0002344
```

```r
plot(model_speed)
```




Q-Q Residuals
Standardized residuals
Theoretical Quantiles
lm(avg_waiting ~ schedalg)
250
249
244


Scale-Location
√|Standardized residuals|
Fitted values
lm(avg_waiting ~ schedalg)
244




```r
Anova(model_speed)
```

```
Anova Table (Type II tests)

Response: avg_waiting
            Sum Sq  Df F value    Pr(>F)
schedalg   6617329   3  6.6443 0.0002344 ***
Residuals 98265708 296
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusions